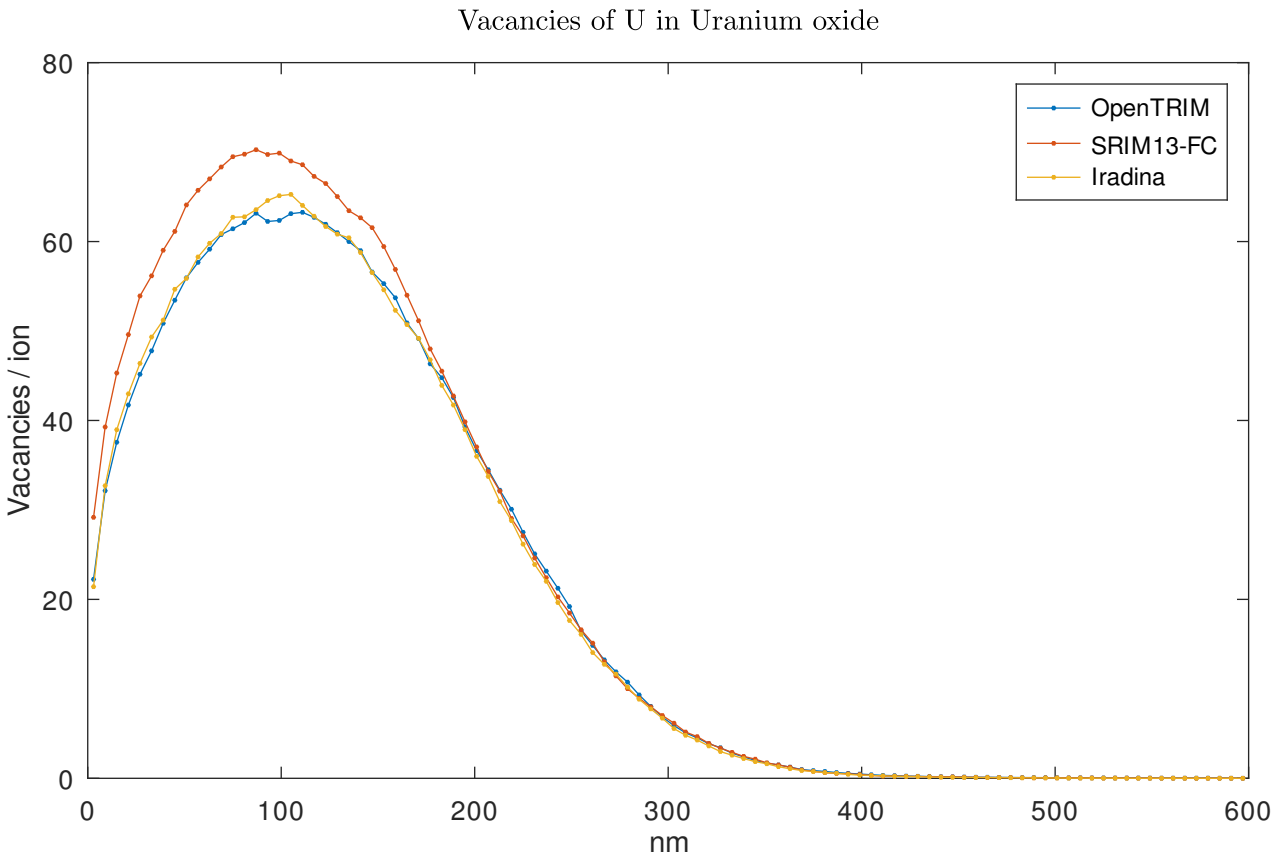
Benchmark #4
1MeV Xe on UO2

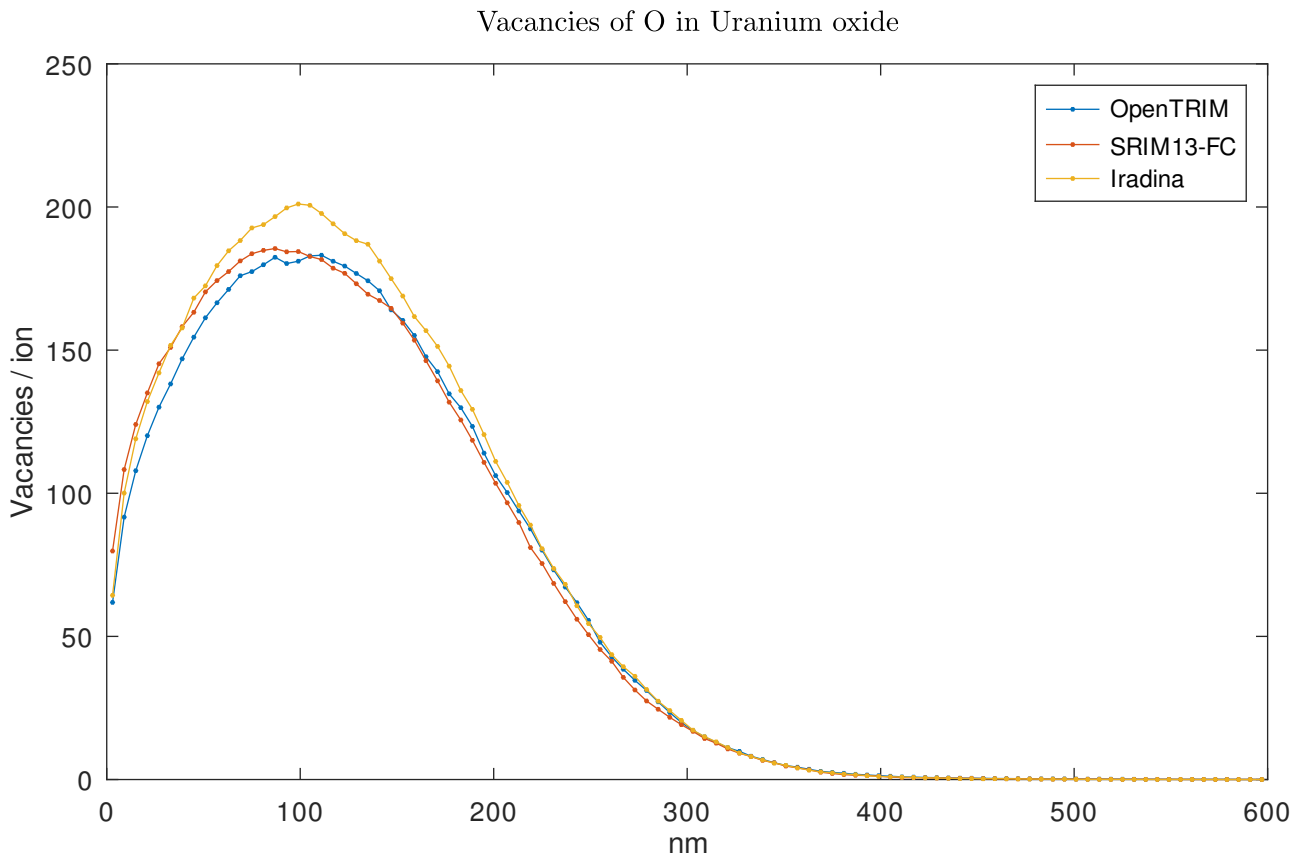
Ion energy E0 = 1e+06 eV

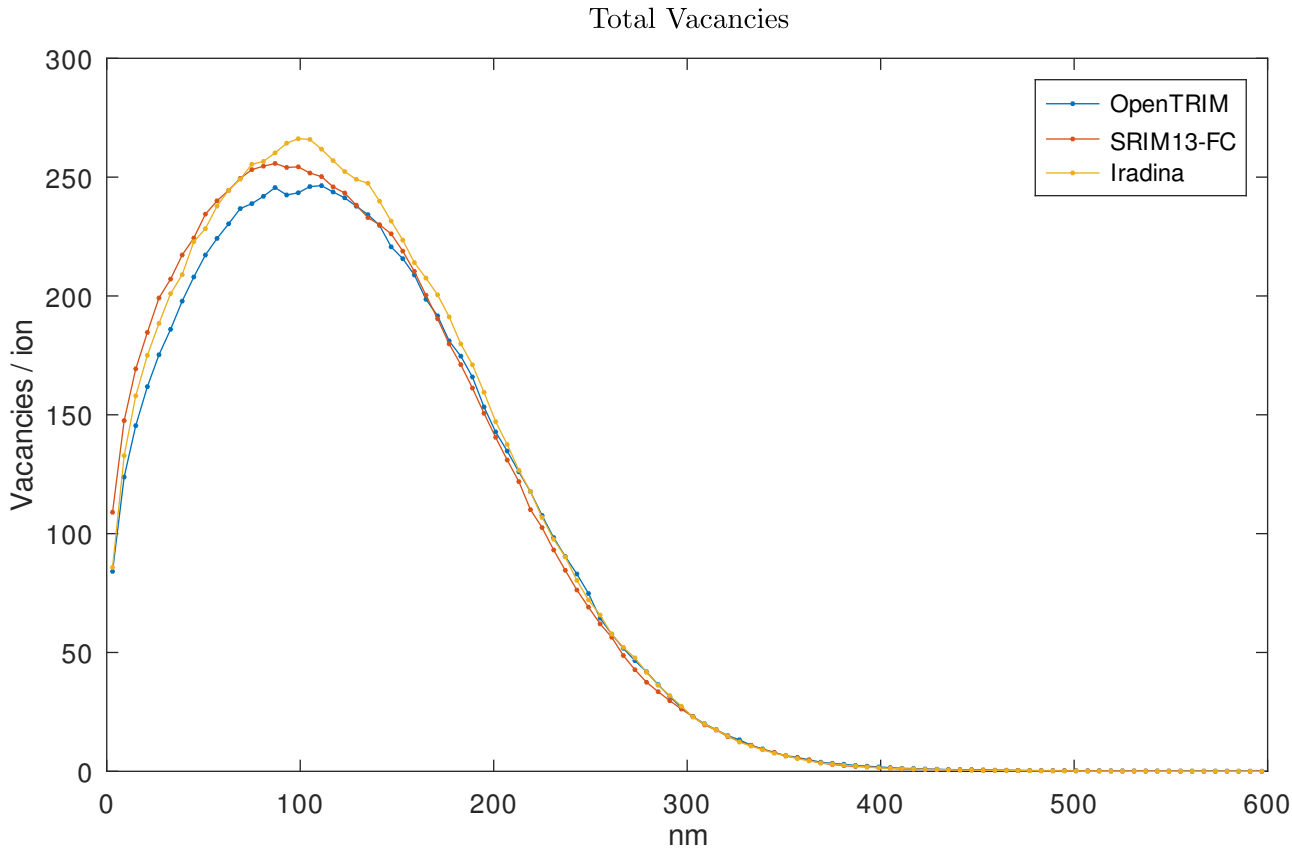
Target depth = 600 nm

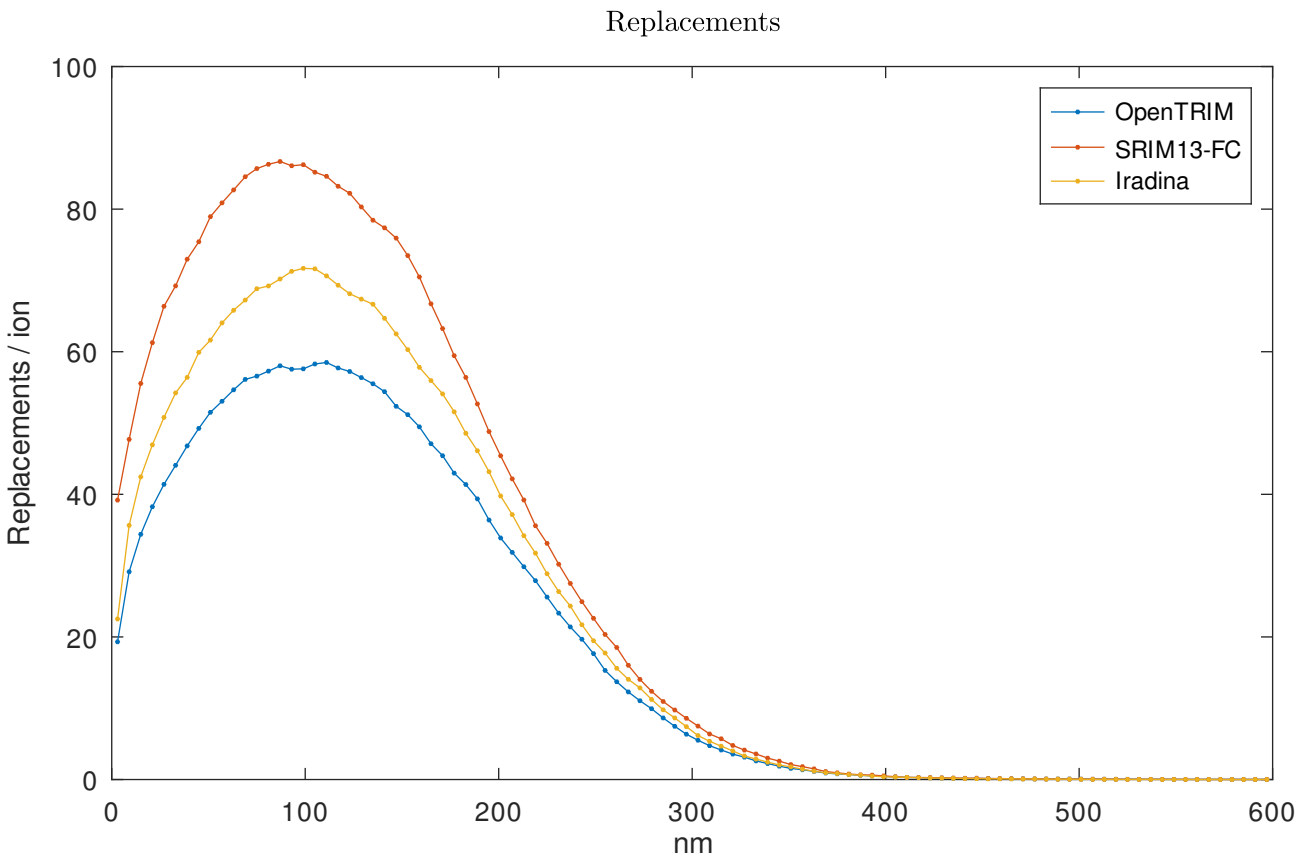
Summary Table

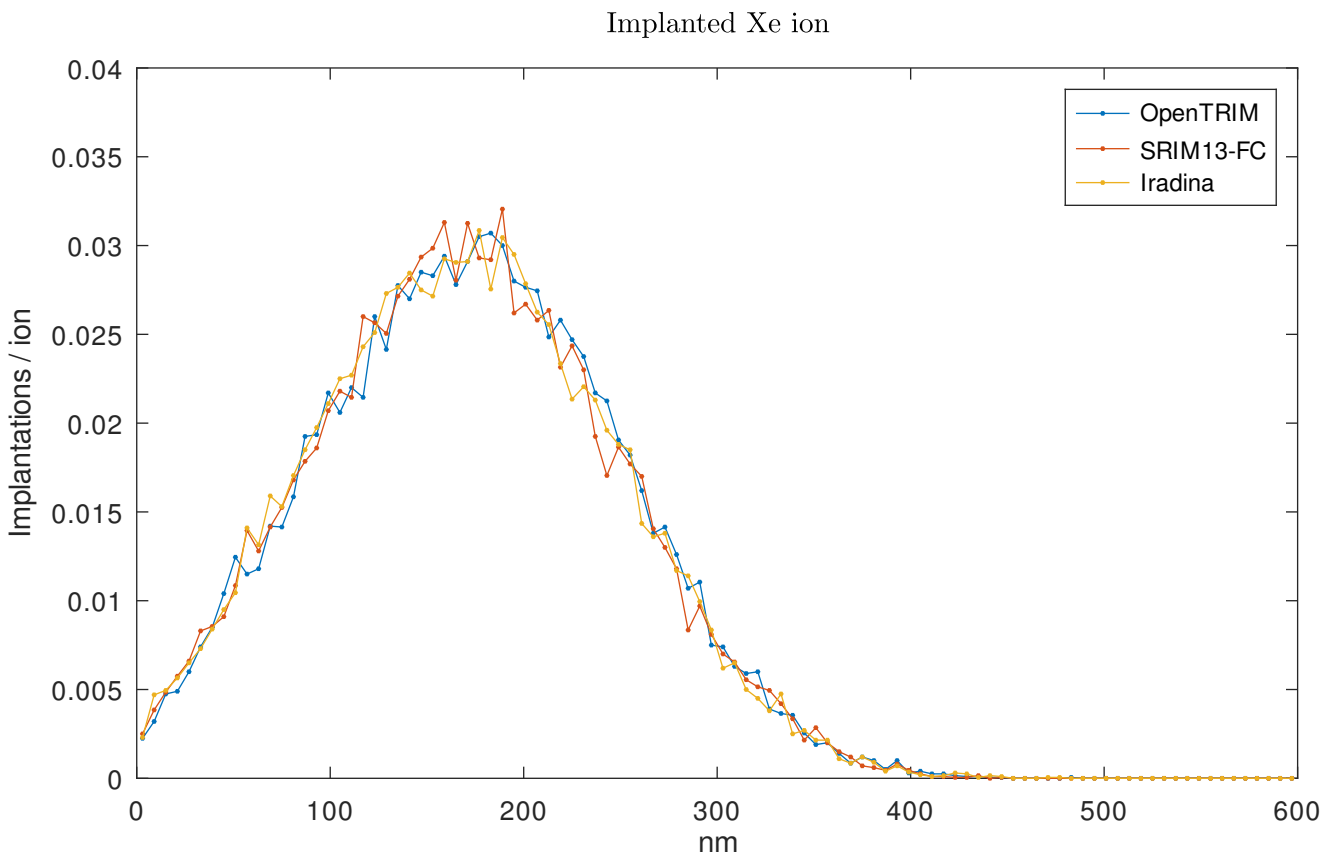
Quantity	OpenTRIM	SRIM13-FC	Iradina
V(U)	2.13e+03	2.31e+03	2.12e+03
V(O)	6.16e+03	6.21e+03	6.55e+03
V(tot)	8.29e+03	8.52e+03	8.68e+03
R(tot)	1.96e+03	2.85e+03	2.34e+03
I(Xe)	0.99	0.986	0.988
EI(Xe)/E0	0.162	0.16	0
EI(r)/E0	0.354	0.346	0
EI/E0	0.516	0.507	0.534
EPh(Xe)/E0	0.00268	0.00241	0
EPh(r)/E0	0.474	0.452	0
EPh(tot)/E0	0.477	0.455	0.458
1-(EI+EPh)/E0	0.00706	0.0388	0.00728

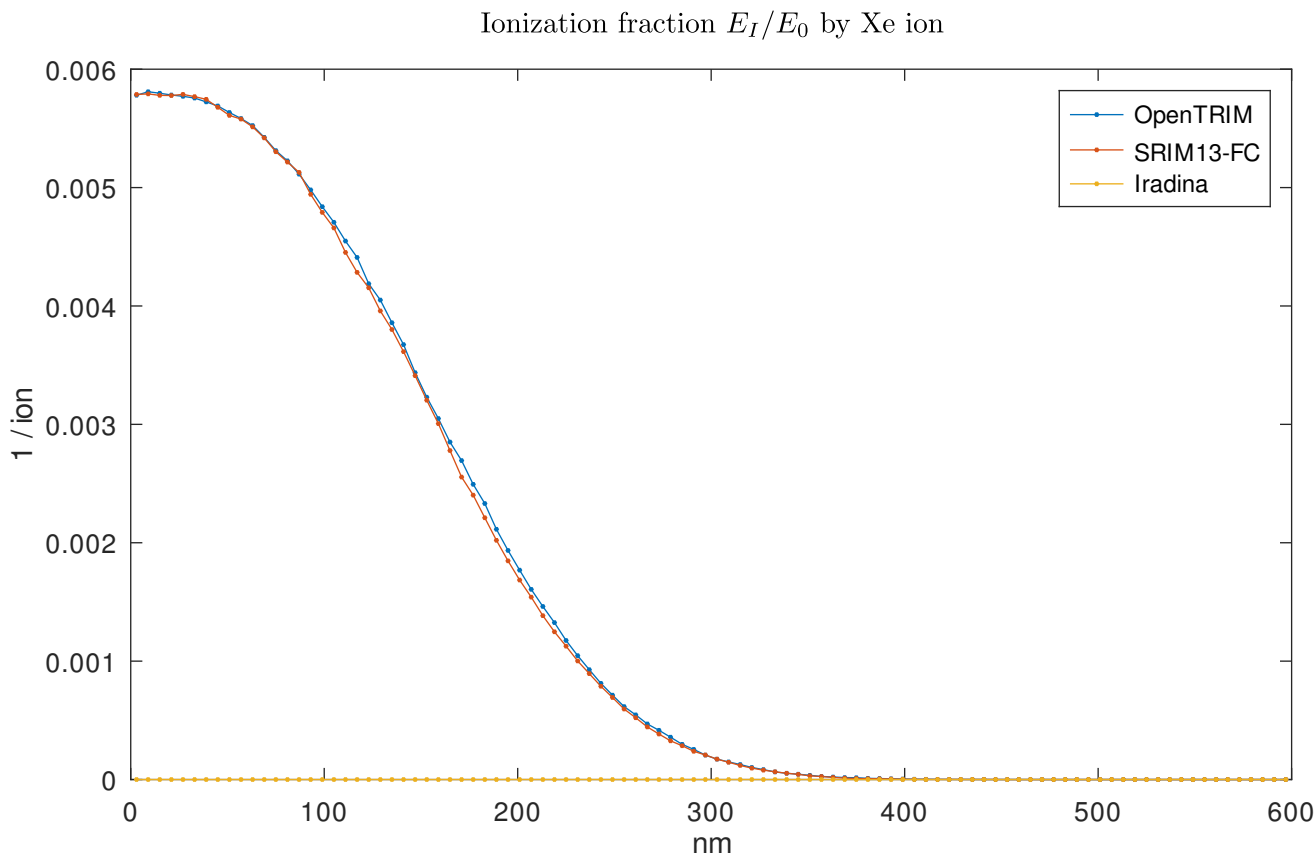

Vacancies of U in Uranium oxide
Vacancies / ion
nm
OpenTRIM
SRIM13-FC
Iradina
0
20
40
60
80
0
100
200
300
400
500
600

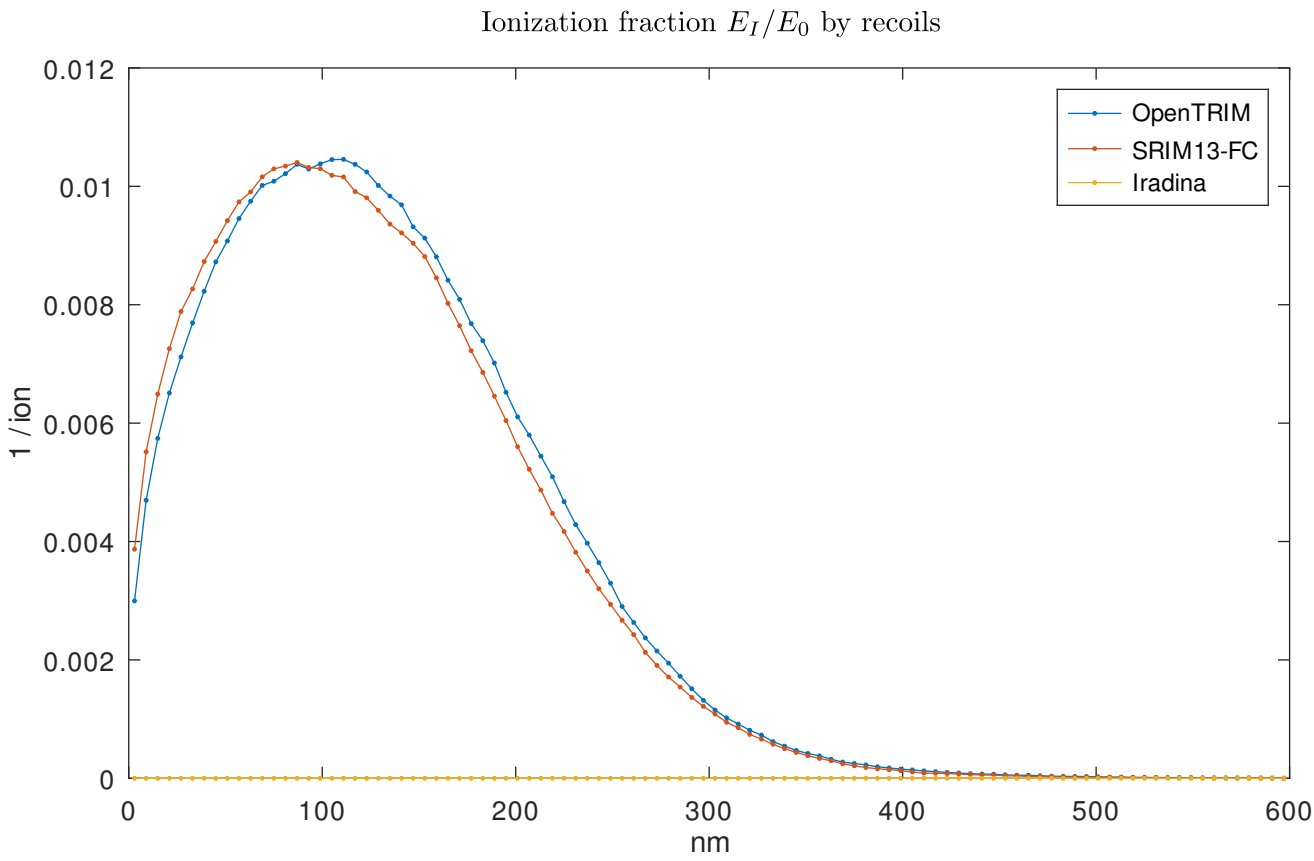

Vacancies of O in Uranium oxide
Vacancies / ion
nm
OpenTRIM
SRIM13-FC
Iradina
0
50
100
150
200
250
0
100
200
300
400
500
600

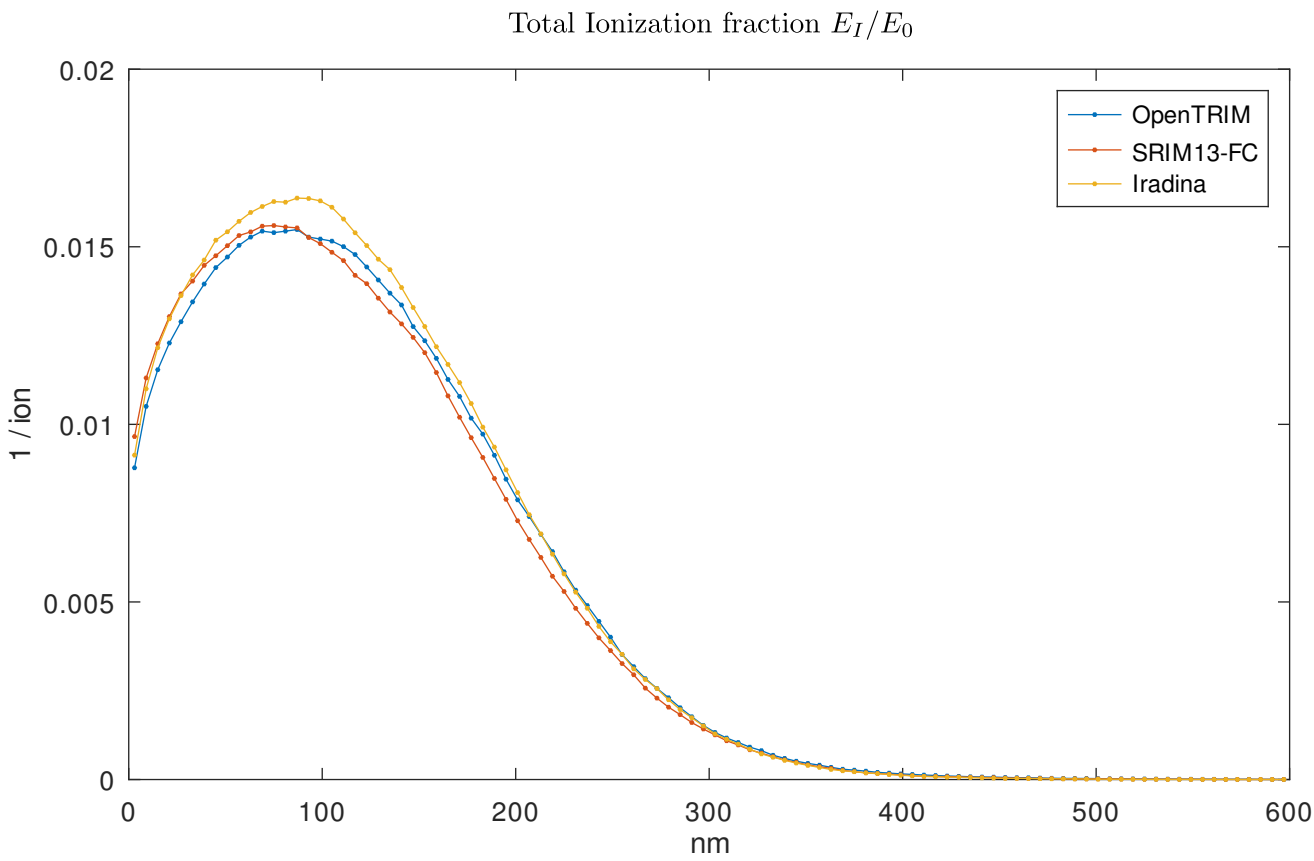

Total Vacancies
Vacancies / ion
nm
OpenTRIM
SRIM13-FC
Iradina
0
50
100
150
200
250
300
0
100
200
300
400
500
600

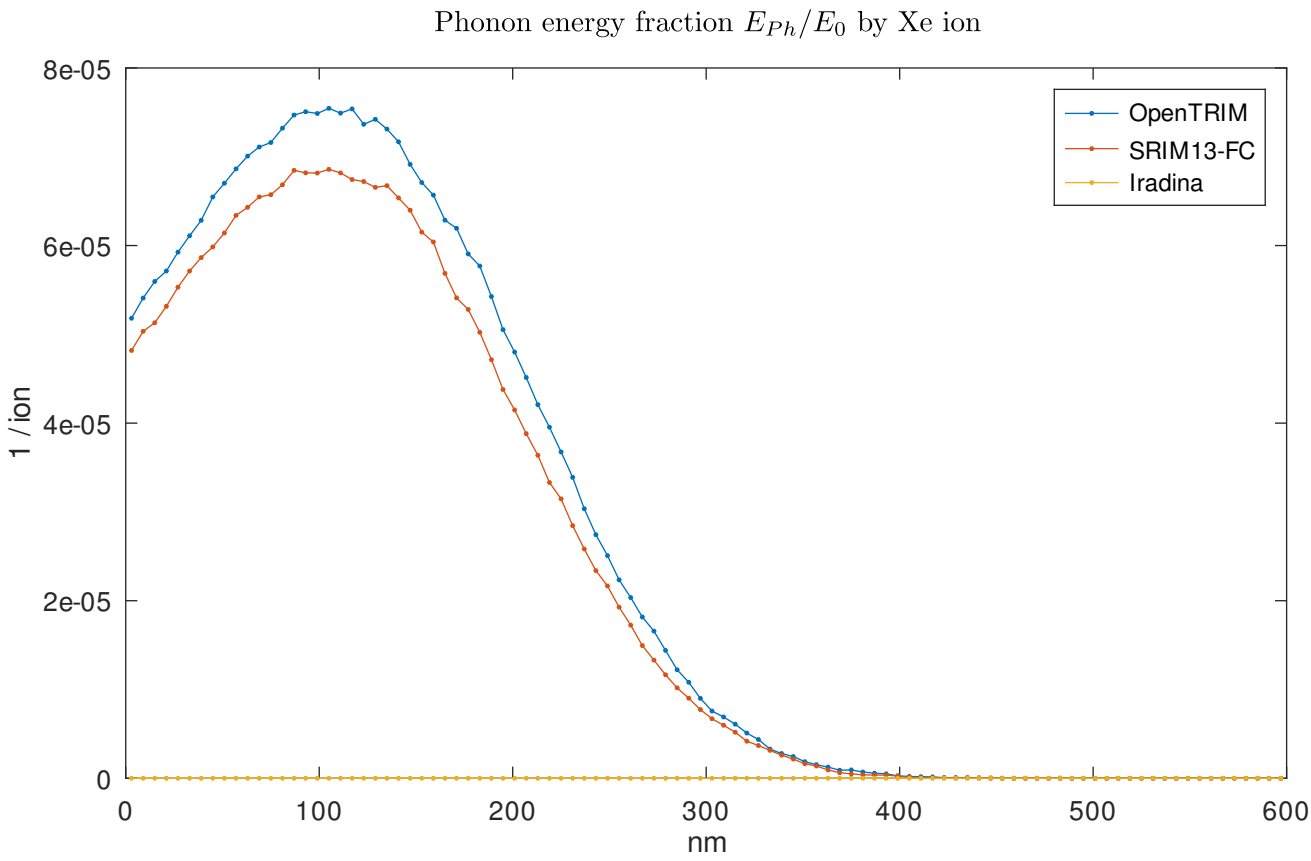

Replacements
Replacements / ion
nm
OpenTRIM
SRIM13-FC
Iradina
0
20
40
60
80
100
0
100
200
300
400
500
600

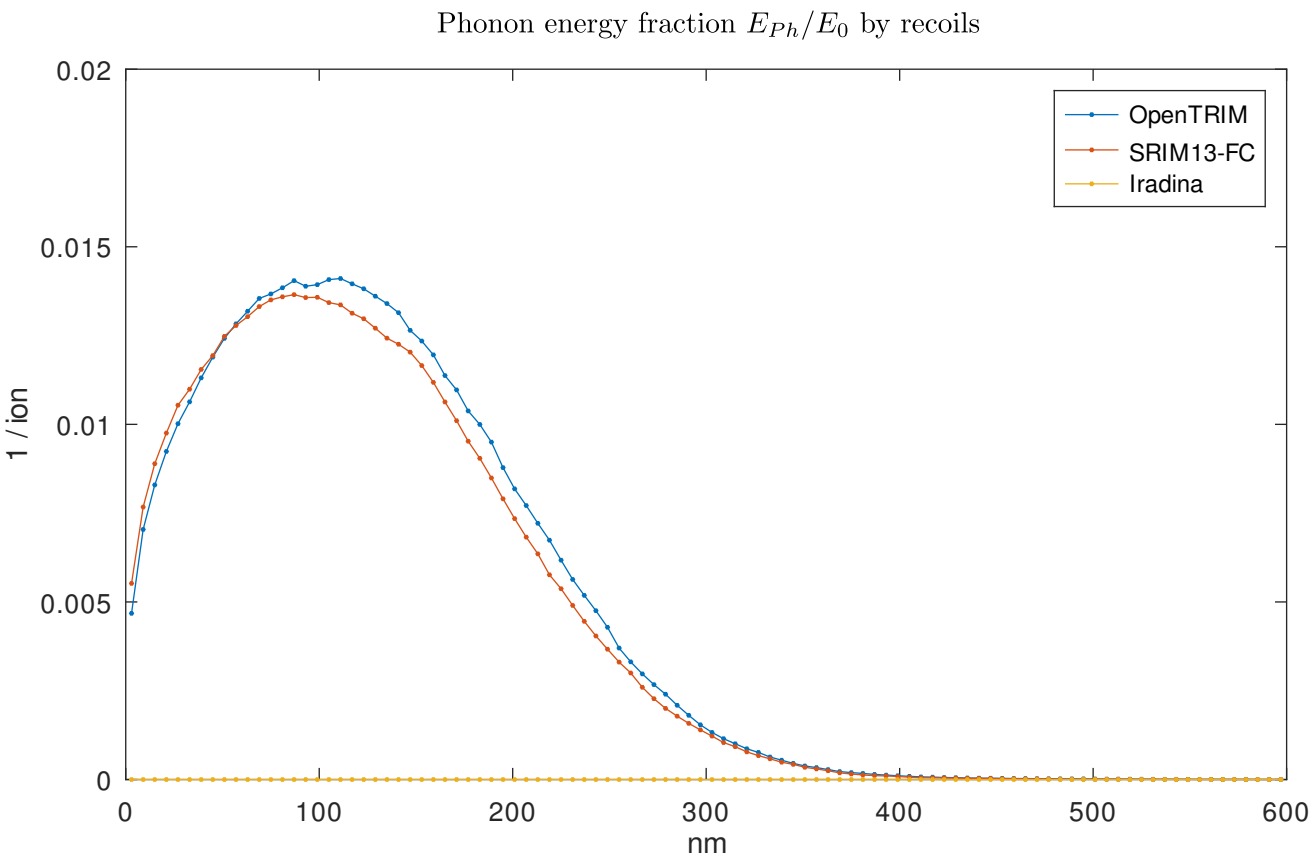

Implanted Xe ion
OpenTRIM
SRIM13-FC
Iradina
Implantations / ion
nm
0.04
0.035
0.03
0.025
0.02
0.015
0.01
0.005
0
0
100
200
300
400
500
600

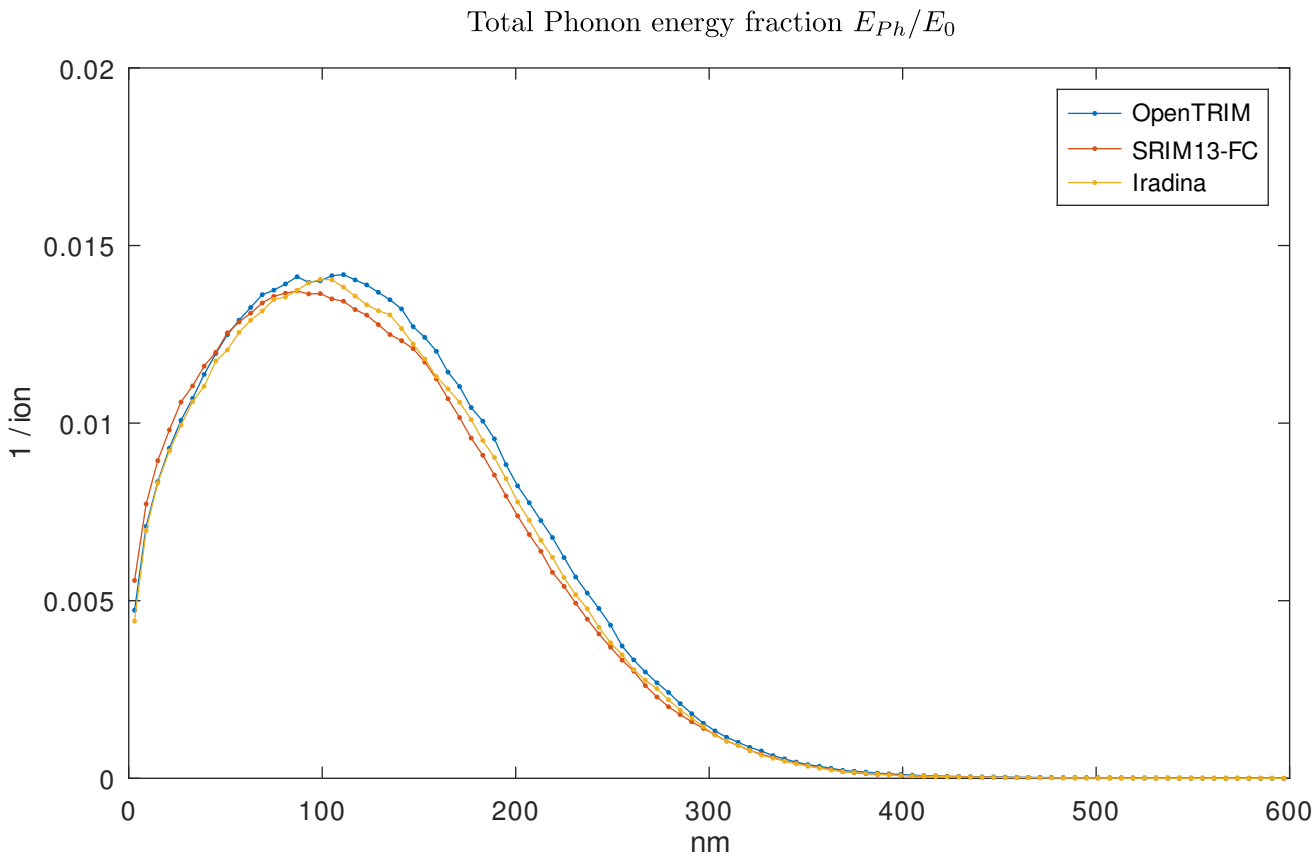

Ionization fraction E_I/E_0 by Xe ion
OpenTRIM
SRIM13-FC
Iradina
1 / ion
nm
0.006
0.005
0.004
0.003
0.002
0.001
0
0
100
200
300
400
500
600

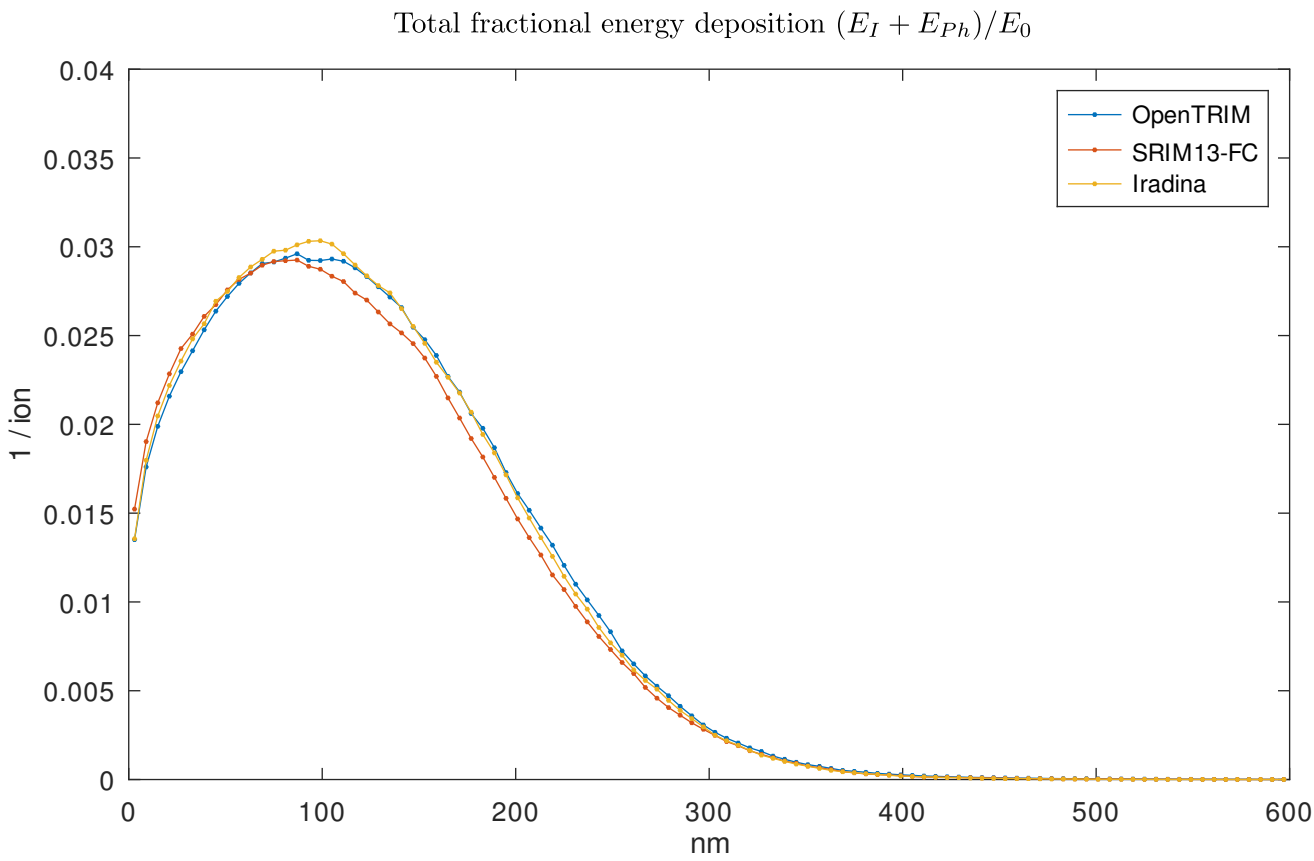

Ionization fraction E_I/E_0 by recoils
0.012
0.01
0.008
0.006
0.004
0.002
0
1 / ion
0
100
200
300
400
500
600
nm
OpenTRIM
SRIM13-FC
Iradina


Total Ionization fraction E_I/E_0
0.02
0.015
0.01
0.005
0
1 / ion
0
100
200
300
400
500
600
nm
OpenTRIM
SRIM13-FC
Iradina


Phonon energy fraction E_{Ph}/E_0 by Xe ion
8e-05
6e-05
4e-05
2e-05
0
1 / ion
0
100
200
300
400
500
600
nm
OpenTRIM
SRIM13-FC
Iradina


Phonon energy fraction E_{Ph}/E_0 by recoils
0.02
0.015
0.01
0.005
0
1 / ion
0
100
200
300
400
500
600
nm
OpenTRIM
SRIM13-FC
Iradina


Total Phonon energy fraction E_{Ph}/E_0
OpenTRIM
SRIM13-FC
Iradina
1 / ion
nm
0.02
0.015
0.01
0.005
0
0
100
200
300
400
500
600


Total fractional energy deposition $(E_I + E_{Ph})/E_0$
OpenTRIM
SRIM13-FC
Iradina
1 / ion
nm
0.04
0.035
0.03
0.025
0.02
0.015
0.01
0.005
0
0
100
200
300
400
500
600